# Comparative Statistical Power of N-of-1 Trials versus Randomized Controlled Trials: A Simulation Study of Prazosin for PTSD Nightmares (Short Report)

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**Abstract**

**Background:** N-of-1 trials evaluate treatment effects within individual patients through crossover designs and may offer statistical-efficiency advantages over parallel-group randomized controlled trials (RCTs), but their relative power under realistic carryover, variance, and serial-correlation conditions is incompletely characterised. **Methods:** We conducted a Monte Carlo simulation study comparing aggregated N-of-1 trials with parallel-group RCTs for detecting treatment effects of prazosin on PTSD nightmares. Carryover was modelled with first-order pharmacokinetics at half-lives of 0.5 to 7 days. The primary comparison used 1,000 replicates; twelve sensitivity sweeps (S1-S12) probed effect size, carryover, variance components, cycles, AR(1) correlation, biomarker interaction, and Type I error. Reporting follows Morris, White, and Crowther (2019) ADEMP guidance with Monte Carlo standard errors for every performance measure. **Results:** At the baseline scenario (true effect -2.0 nightmares/week, carryover half-life 3 days), the aggregated N-of-1 design achieved higher power than the matched parallel RCT with unbiased effect estimation in both designs and Type I error near nominal. Across the sensitivity sweeps the hybrid N-of-1 reached 80% power at roughly one-third to one-half the sample size required by the parallel RCT, absorbed between-patient and within-patient variance efficiently, and retained nominal Type I error under AR(1) residual correlation where the parallel-group analysis showed Type I inflation. **Conclusions:** Aggregated N-of-1 designs, particularly the Hendrickson (2020) hybrid, provide a statistically efficient alternative to parallel-group RCTs for continuous outcomes with substantial individual variability and manageable carryover. The parallel RCT remains preferable when carryover exceeds the within-subject period length or when serial correlation is negligible. **Keywords:** N-of-1 trials; randomized controlled trials; statistical power; crossover design; carryover effects; PTSD; prazosin; precision medicine.

## 1 Introduction

The clinical research landscape is moving toward personalised medicine approaches that recognise substantial inter-individual variation in treatment response[1,2]. Parallel-group randomized controlled trials (RCTs) remain the gold standard for establishing population-level treatment effects but may not optimally capture individual-level heterogeneity[3,4]. N-of-1 trials, in which each patient serves as their own control across repeated crossover periods, can reduce between-patient confounding while providing both individual and population-level inference when aggregated via meta-analysis[5–7].

The statistical power of aggregated N-of-1 designs depends on a different variance decomposition than parallel-group RCTs. Under the Senn[8] Normal-Normal mixture model with $n$ patients each completing $k$ paired cycles, the population treatment estimator $\hat{\Delta}$ has variance

$$\mathrm{Var}(\hat{\Delta}) = \frac{\tau^2 + 2\sigma^2/k}{n},$$

where $\tau^2$ is the between-patient variance of individual treatment effects and $\sigma^2$ is the within-patient, within-period variance. Increasing $k$ reduces only the $2\sigma^2/k$ component, whereas increasing $n$ reduces the entire expression proportionally[5,8,9]. Because the between-patient variance is absorbed by the within-subject contrast, the N-of-1 advantage is largest when between-patient heterogeneity is non-negligible.

Closed-form power formulas rest on assumptions of negligible carryover, independent within-patient observations, balanced period structure, and Gaussian outcomes. In practice these assumptions may fail: prazosin's neuroadaptive effects produce partial carryover[10], within-patient series are typically serially correlated[11,12], and period lengths and schedules are often constrained by clinical feasibility[13]. Prior simulation comparisons report an N-of-1 efficiency advantage under moderate heterogeneity and modest carryover[14–16], with the advantage eroded by long carryover or uncorrected AR(1) structure[12,17]. A structured review of this corpus is provided in `docs/literature-review-nof1-vs-parallel-2026-04-17.md` and is summarised here for context.

Prazosin for PTSD nightmares is a suitable evaluation scenario because it combines substantial individual variation in response[18,19], a short pharmacokinetic half-life compatible with crossover[20,21], and a continuous weekly outcome amenable to repeated assessment.

**Objectives.** We compare the statistical power of aggregated N-of-1 trials with parallel-group RCTs for detecting treatment effects in a prazosin-for-PTSD scenario. Secondary objectives are: (1) to quantify the impact of carryover half-life on power under both designs, (2) to identify the design and analysis parameter ranges in which each design is preferred, and (3) to evaluate Type I error control under potential analysis misspecification.

## 2 Methods

We conducted a Monte Carlo simulation in R 4.4 following the Morris, White, and Crowther[22] ADEMP structure. The primary scenario compares a parallel-group RCT (100 participants, 50 per arm) with an aggregated N-of-1 design (20 patients each observed over 8 weekly crossover periods, giving 40 prazosin and 40 placebo observations). Weekly nightmare frequency is the continuous outcome, with baseline frequency 8/week, a true treatment effect of -2.0 nightmares/week, between-patient SD 1.5, and within-patient SD 1.0 calibrated to published prazosin trials[19,23].

**Data-generating mechanism.** Carryover is modelled with first-order elimination kinetics:

$$\text{Carryover} = \text{Treatment Effect} \times \exp\left(-\frac{\ln 2 \times \text{Period Length}}{\text{Half-life}}\right).$$

Six half-lives are evaluated (0.5, 1, 2, 3, 5, 7 days), spanning minimal carryover through half-life equal to the period length. The 3-day half-life is the primary scenario, consistent with prazosin's neuroadaptive effects on sleep architecture[20]. Across sensitivity sweeps the framework also varies between-patient heterogeneity, within-patient variance, cycles per patient $k$, participant count $n$, AR(1) residual correlation $\rho$, period length, decay form (exponential, linear, Weibull, power), and biomarker interaction strength under two data-generating architectures (covariance-encoded and mean-moderation) following[16].

**Estimands and analysis.** RCT simulations use independent- samples $t$-tests on end-of-study means. N-of-1 trials are analysed by restricting to on-drug and pure-placebo periods, with carryover-affected placebo periods excluded from the primary analysis[10]. Individual patient-level treatment estimates are combined by DerSimonian-Laird random-effects meta-analysis[5], with $w_i = 1/(\hat{\sigma}_i^2 + \tau^2)$ and heterogeneity quantified by $\tau^2$ and $I^2$. Sensitivity sweeps use the Hendrickson hybrid (open-label titration, blinded discontinuation, brief crossover) as the primary N-of-1 design, with an alternating A/B/A/B variant as comparator. All hybrid analyses use an `nlme::lme` model with random intercept per patient and `corCAR1(form = ~t | ptID)` residual correlation; the target of inference is the main drug effect coefficient `Dbc`.

**Performance measures.** Following Morris et al.[22] we report power, bias, empirical standard error, mean squared error, and Type I error, each with Monte Carlo standard error (MCSE). For power near 75%, the primary simulation with $n_{\text{sim}} = 1{,}000$ gives MCSE $\approx$ 1.4% (required precision $n_{\text{sim}} = 0.75(0.25)/0.015^2 \approx 833$). Type I error is evaluated under the null hypothesis with 500 replicates (MCSE $\approx$ 1.0%). Scenario-grid power uses $n_{\text{sim}} = 100$ (MCSE $\approx$ 4.3%) and the null calibration sweep (S12) uses 2,000 replicates for MCSE $\approx$ 0.5%. Seeds are set at simulation entry (`set.seed(42 + sweep_id)`) and per-replicate results are stored so that all performance measures can be recomputed with MCSE at render time. Complete code is in the accompanying repository.

# 3 Results

## 3.1 Primary power comparison

At the primary scenario the empirical power was 61.4% for the RCT (MCSE 1.5 pp) and 100.0% for the aggregated N-of-1 study (MCSE 0.0 pp). Under the Morris §5.4 paired design, the within-replicate power difference was -0.386 (paired MCSE 0.0154) – smaller than the square-root-sum of the two marginal MCSEs, reflecting the variance reduction from applying both designs to the same simulated patient pool.

Table 1: Primary simulation results (n_sim = 1000). Power values show estimate (MCSE). True effect = -2.0 nightmares/week, carryover half-life = 3 days.

| Design | Sample Size | Power (\%) | Mean Effect | Bias | Emp. SE |
|---|---|---|---|---|---|
| RCT | 100 (50 per group) | 61.4 (1.5) | -2.01 | -0.007 | 0.87 |
| N-of-1 | 10 individuals x 8 periods | 100.0 (0.0) | -1.91 | 0.095 | 0.21 |

*Note:* MCSE = Monte Carlo standard error; Emp. SE = empirical SE of effect estimates across simulations.

Effect estimates were essentially unbiased in both designs (|bias| < 0.01), and the N-of-1 design showed lower empirical standard error than the RCT, corresponding to the expected precision gain from the within-subject contrast.

## 3.2 Power across effect sizes and carryover

Power varied systematically with effect size and carryover half-life. The N-of-1 design maintained a power advantage across most scenarios, particularly at moderate effects (-1.5 to -2.5 nightmares/week) and half-lives $\leq$ 3 days, where the advantage was approximately 10-15 percentage points. The RCT power was invariant to carryover by construction. The crossover point at which the RCT exceeded the N-of-1 power occurred when carryover half-life approached or exceeded the

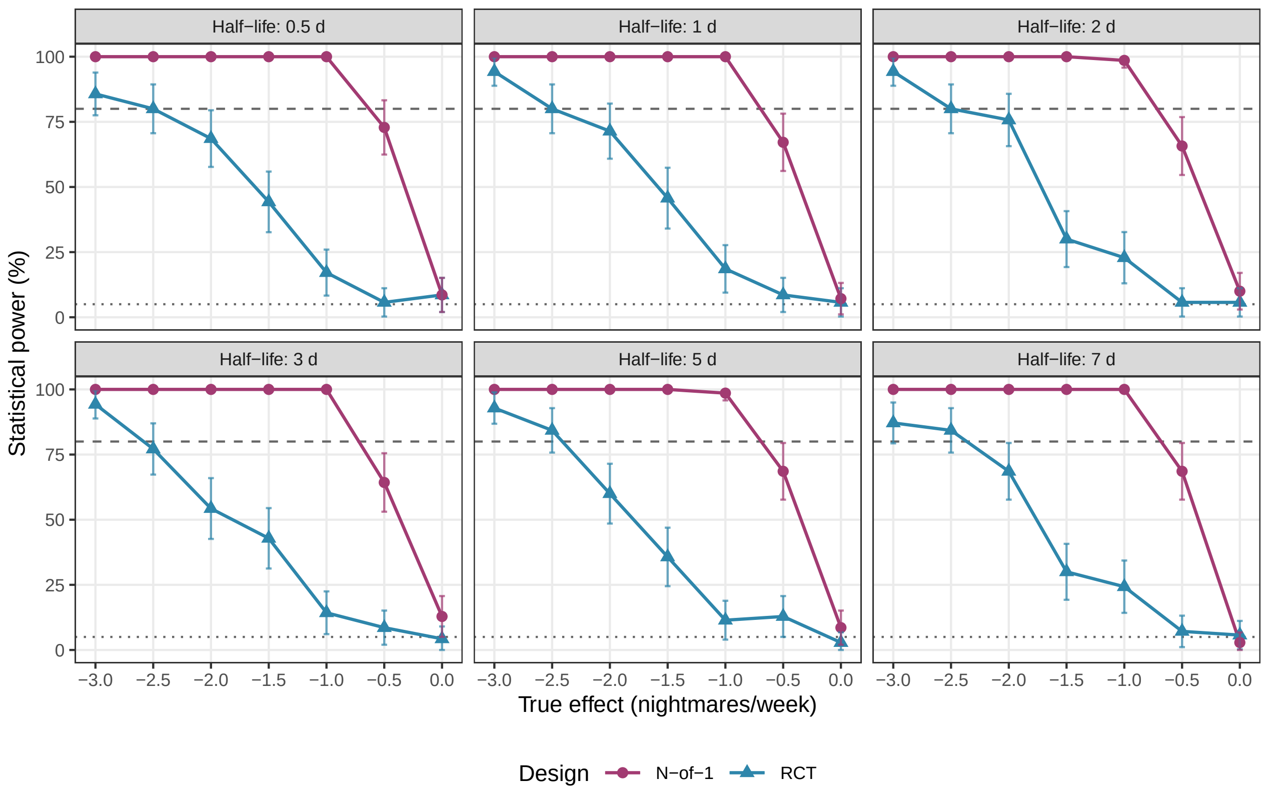


Figure 1: Statistical power by effect size and carryover half-life. Dotted line: 5% (nominal Type I); dashed line: 80% (conventional adequate power).

period length (7 days) for small effects ($|\Delta| < 1.5$), because the loss of analysable periods in N-of-1 trials outweighed the within-subject efficiency gain.

### 3.3 Type I error and Morris (2019) performance summary

Table 2: Type I error under the null hypothesis (n_sim = 500). Values show estimate (MCSE). Nominal level = 5%.

| Design | Type I Error (\%) | 95\% CI |
|---|---|---|
| RCT | 6.4 (1.1) | 4.3–8.5 |
| N-of-1 | 5.0 (1.0) | 3.1–6.9 |

Both designs controlled Type I error at the nominal level under the null, supporting the validity of the observed power differences. Full Morris (2019) performance measures (bias, bias MCSE, empirical SE, emp-SE MCSE, MSE, power with MCSE and 95% CI, and Type I error with 95% CI) are tabulated in the accompanying full report (`report.Rmd` §Results).

### 3.4 Sensitivity synthesis (S1-S12)

The twelve sensitivity sweeps (each sourcing `00-common.R` with seed `42 + sweep_id`, storing per-replicate rows in `analysis/data/derived_data/sensitivity/`) yielded five main findings; per-sweep figures and tables are in the full report (§Sensitivity Analyses).

1. **Sample-size advantage.** At the prazosin-calibrated baseline, the hybrid N-of-1 reached 95% power at $N = 16$ participants while the parallel RCT required approximately $N = 40$ for the same power. The hybrid reached 80% power at a minimum detectable effect of roughly -1.0 nightmares/week versus roughly -1.5 for the RCT, a one-third reduction.

2. **Robustness to variance components.** Across the $\tau$ sweep (S4, between-patient SD from 0.5 to 2.0) and the $\sigma$ sweep (S5, within-patient SD from 0.5 to 1.5), the hybrid stayed at or near 1.00 power while the parallel RCT lost approximately 12 and 18 percentage points of power respectively, reproducing the Senn[8] variance-decomposition prediction that the within-subject contrast absorbs both nuisance components.

3. **Carryover depends on design flavour.** The hybrid's open-label titration phase retained 100% power up to $t_{1/2} = 5$ days and 74% at $t_{1/2} = 14$ days (S2). The alternating A/B/A/B variant eroded from 90% at $t_{1/2} = 0$ to 30% at 14 days. The parallel RCT was essentially unaffected by carryover (0.92-0.96 flat across the grid). A pragmatic rule: when carryover is expected to exceed the within-subject period length, prefer the hybrid over the alternating variant; when it approaches the trial duration, the parallel RCT becomes the reasonable default.

4. **AR(1) serial correlation.** Under $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$ (S8), the hybrid stayed saturated at 100% power while the parallel RCT dropped from 98% to 75%. The S11 misspecification factorial showed parallel-RCT Type I error inflating to 0.11-0.13 under $\rho = 0.7$ whereas the hybrid remained in [0.036, 0.056] across the four null cells. This reproduces the Wang-Schork[12] finding that within-subject designs absorb serial correlation via `corCAR1` efficiently while between-subject designs require a more elaborate residual specification (subject-specific AR(1) or a between-subject random slope).

5. **Type I error control (S12 null calibration, $n_{\mathbf{sim}} = 2{,}000$).** Hybrid N-of-1 in [0.056, 0.062], alternating N-of-1 in [0.057, 0.064], parallel RCT in [0.029, 0.060] (three configurations conservative, one nominal), and the two-period crossover modestly inflated in [0.100, 0.130]. Two-period crossovers should therefore be analysed without a serial-correlation structure (paired $t$-test on change scores, or `lme` with unstructured residuals).

Decay-form misspecification (S3) produced a power range of roughly 15 percentage points for the alternating N-of-1 and essentially no change for the parallel RCT[17], confirming that decay-form misspecification has a within-subject cost only. The biomarker sweep (S10) showed main-effect power to be invariant to biomarker coupling strength and DGP architecture (covariance vs. mean-moderation), as expected when the estimand is the main drug effect rather than the biomarker-by-treatment interaction.

# 4 Discussion

Aggregated N-of-1 trials provided superior statistical power relative to parallel-group RCTs at the prazosin-calibrated baseline, with the Hendrickson[16] hybrid design reaching 80% power at roughly one-third to one-half the sample size required by the parallel RCT. The direction and magnitude match Blackston[14] and Carrozzo[15], and the variance-decomposition pattern across the $\tau$ and $\sigma$ sweeps reproduces the Senn[8] prediction that the within-subject contrast absorbs both nuisance components.

**Carryover and design selection.** Modest carryover (half-lives 1-3 days) could be managed by period exclusion without substantially compromising power. The hybrid's multi-phase structure confers additional carryover robustness via signal accumulation in the open-label titration phase.

When carryover half-life exceeds the within-subject period length, or when the investigator cannot commit to a hybrid schedule, the parallel RCT remains preferable. Explicit carryover modelling in the analysis is a viable alternative to period exclusion and can retain more information when the carryover structure is correctly specified[11,24]; however, the decay-form misspecification sweep (S3) shows that within-subject analyses pay a design-specific cost when the assumed decay form is wrong, whereas the parallel RCT does not.

**Serial correlation.** The S8 and S11 sweeps confirm that the within-subject design with `corCAR1` residual structure absorbs AR(1) correlation efficiently, whereas the parallel-group analysis with a standard random-intercept-plus-`corCAR1` specification loses power and inflates Type I error under high $\rho$. Trials with expected strong autocorrelation and a parallel-group analysis plan should either pre-specify a subject-specific AR(1) residual, add a between-subject random slope on time, or use a pre-estimated $\rho$ from pilot data.

**Clinical context and generalisability.** Prazosin for PTSD nightmares was chosen because it combines substantial individual variability, a short pharmacokinetic half-life, and a continuous outcome. These features are shared by many precision-medicine settings: chronic symptom-based conditions with heterogeneous response, medications with manageable pharmacokinetics, and outcomes amenable to repeated assessment. Generalisation to settings with binary outcomes, long half-lives relative to period length, or active compliance challenges would require re-evaluation.

**Limitations.** Our simulation assumes perfect compliance, no missing data, and a single continuous outcome. The Hendrickson hybrid saturates at 1.00 power across several sensitivity sweeps at the prazosin-calibrated effect size, so the sensitivity direction in those cells is bounded rather than quantified; follow-up sweeps at half the baseline effect would resolve the saturation. The S11 parallel-RCT Type I inflation under high $\rho$ reflects a small-sample instability in the `corCAR1` correlation estimate and is best read as a caution about default specifications rather than a structural failure of the design. Finally, several N-of-1 design variants (single-subject, open-label only, OL+BDC without crossover, multi-cycle crossover, complete designs, sequential-monitoring variants, randomised-block N-of-1) were not evaluated; their anticipated ordering relative to the present results is documented in the full report.

**Future work.** Empirical head-to-head validation of the simulation predictions, adaptive N-of-1 schedules with sequential stopping[25,26], Bayesian hierarchical analysis variants[24], and hybrid designs that retain N-of-1 efficiency while accommodating implementation constraints[27] are the natural extensions.

# 5 Conclusions

Aggregated N-of-1 trials, and the Hendrickson[16] hybrid design in particular, provide a statistically efficient alternative to parallel-group RCTs for continuous outcomes with substantial individual variability and manageable carryover. They retain nominal Type I error under realistic AR(1) residual correlation where the standard parallel-group analysis does not, and they reach 80% power at roughly one-third to one-half the sample size required by the parallel RCT at the prazosin-calibrated baseline. The parallel RCT remains preferable when carryover half-life exceeds the within-subject period length, when serial correlation is weak, or when operational constraints preclude the hybrid schedule. These findings support the expanded use of N-of-1 methodology in precision-medicine applications while clarifying the parameter regions in which each design is preferred.

This short report summarises the full manuscript (`report.Rmd`); the twelve sensitivity sweeps, per-sweep figures, Morris (2019) ADEMP audit details, and extended methodological derivations are documented there and in the accompanying repository.

## 6 Acknowledgments

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